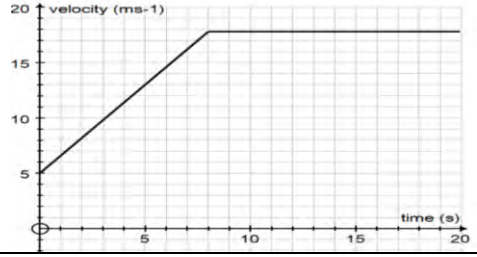
Question			Answer	Marks	AOs		Guidance
14	(i)		$u=5,\ v=11.4,\ t=4$ $a=\frac{v-u}{t}=\frac{11.4-5}{4}=1.6$ $v=5+1.6t$	**M1** **A1** **A1** **[3]**	**3.1b** **1.1b** **3.3**	Using *suvat* equation(s) leading to value for a Any form FT their a	
	(ii)		The car would not be able to accelerate indefinitely – the velocity would become too large	**E1** **[1]**	**3.5b**		
	(iii)		When $v=17.8$ $t=\frac{17.8-5}{1.6}=8$  	**B1** **G1** **G1** **[3]**	**1.1a** **1.1a** **3.5c**	Calculation or point on graph labelled at $t = 8$ Two line segments with one horizontal Axes labelled. (0, 5) and constant speed 17.8 clear on vertical scale	Mark intent for 17.8 – allow for a linear scale beyond 17.8
	(iv)		Dividing area into sections Area under trapezium = $\frac{1}{2}(5+17.8)\times 8=91.2$ Area rectangle $12\times 17.8=213.6$ Total displacement = 304.8 m	**M1** **A1** **A1** **[3]**	**3.1b** **1.1a** **1.1b**	 May be found as sum of areas. May be implied by correct total FT their distance found for first 8s	FT their graph if linear for M1 A0 for a triangle or trapezium area 213.6 must be added to another distance
	(v)		When $t=4\ \ v=5+0.3\times 4^2-0.05\times 4^3=11.4\ \text{m s}^{-1}$ Which matches the given value	**B1** **[1]**	**3.4**	Allow without comment	
	(vi)		$\frac{dv}{dt}=0.6\times 2t-0.05\times 3t^2\ \left[=1.2t-0.15t^2\right]$ When $t=8\ \ v=1.2\times 8-0.15\times 64=0$ Acceleration is zero at $t=8$ which means that the car reaches its maximum speed without the sudden change in acceleration in model A.	**M1** **A1** **E1** **[3]**	**1.1a** **3.2a** **3.2a**	Need not be simplified Must mention acceleration Must compare with model A	Final mark can be awarded independently for a statement about change in acceleration as long as supported by some numerical evidence

Question			Answer	Marks	AOs		Guidance
	(vii)		EITHER $\int_0^8 \left(5+0.6t^2-0.05t^3\right)\,\mathrm{d}t = \left[5t+0.2t^3-0.0125t^4\right]_0^8$ =91.2 m which is same as model A for the first 8 s Distance is the same for the remainder of the time So this is the same as model A at $t = 20$	**M1** **A1** **E1** **[3]**	**2.1** **1.1b** **2.1**	 **BC** Must consider to $t = 20$	Allow for correct definite integral stated and calculator used. Also allow M1A1 for $5\times8-0.2\times8^3-0.0125\times8^4$ seen
			OR $\int_0^8 \left(5+0.6t^2-0.05t^3\right)\,\mathrm{d}t = \left[5t+0.2t^3-0.0125t^4\right]_0^8$ =91.2 m Distance at 17.8 ms^{-1} 213.6 Total distance 304.8m [which is the same as model A]	**M1** **A1** **A1**	 	 **BC** Must consider to $t = 20$	Allow for correct definite integral stated and calculator used.